



WORKSTREAM A

Speech Technology for Nigerian Languages in Safety-Critical Voice Applications:

A Systematic Mapping Study of Datasets, Models and Telephony Readiness

Comprehensive Research Plan & Protocol

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This research plan outlines a comprehensive, structured approach for Workstream A to conduct the systematic mapping study on speech technology for Nigerian languages, strictly adhering to the requirements of the project brief.

Phase 1: Preparation & Protocol Development

PHASE OBJECTIVE

Define the exact scope, finalize the methodology, and publicly commit to it to ensure scientific rigor and independence.

1. Define Research Questions & Scope:

- *Core Question:* What ASR and TTS resources exist for Yorùbá, Hausa, Igbo, Nigerian Pidgin, and Nigerian-accented English?
- *Sub-questions:* How do these models perform under 8 kHz narrowband telephony compression? Are non-WER metrics being used for safety-critical evaluations? How are regional dialects handled? What exists for medical/clinical vocabulary? What is the state of Nigerian Pidgin resources?
- *Inclusions:* The resources to be included are: ASR, TTS, language Identification, code-switching detection, dialect handling for the target languages. Datasets, architectures, metrics, telephony robustness.
- *Exclusions:* Machine translation (unless as a voice intermediary), non-Nigerian languages (except as baselines), and clinical content correctness.

2. Develop the Search Strategy:

- *Keywords:* Build Boolean search strings combining language terms (Yoruba, Hausa, Igbo, Pidgin, Naija) with technology terms (ASR, TTS, Speech Recognition, Speech Synthesis, NLP, Narrowband, Telephony).
- *Databases:* Define the search venues: ACL Anthology, IEEE Xplore, ACM Digital Library, PubMed (for clinical NLP crossover), arXiv (preprints), and Hugging Face (for datasets/models not formally published).

3. Establish Inclusion/Exclusion Criteria:

- *Inclusion:* Peer-reviewed papers (last 5 years), seminal historical papers, studies focused on West African settings, and vital preprints (clearly labeled).
- *Exclusion:* Vendor blogs, marketing materials, predatory journals, opinion pieces, and papers evaluated solely on abstracts.

4. Register the Protocol:

- Draft the scoping review protocol using the PRISMA-ScR guidelines.
- Upload and register the protocol on the Open Science Framework (OSF) to preemptively validate the methodology before screening begins.

Phase 2: Literature Search & Screening

PHASE OBJECTIVE

Systematically gather all relevant literature and filter it down to the final included set using a double-blind screening process.

1. Execute the Search:

- Run the finalized search strings across all chosen databases.
- Import all results into a reference manager (e.g., Zotero or Mendeley).
- Remove duplicate entries.

CRUCIAL STEP

Record the exact number of results returned at every stage for the PRISMA-ScR flow diagram.

2. Title and Abstract Screening (Independent, Two Reviewers):

- Two team members will independently review the titles and abstracts against the inclusion/exclusion criteria.
- Any disagreements must be logged and resolved via discussion or a third reviewer.

RULE CHECK

Do not exclude a paper just because it is paywalled. Send the title to Mariam to acquire the full text.

3. Full-Text Review & Quality Assessment:

- Read the full text of all papers that pass the initial screen.

- Assess quality based on the brief’s criteria: look for data leakage, class imbalance, lack of baseline comparisons, and over-reliance on simple accuracy.
- Evaluate the “deployment realism” (e.g., was it tested on a 30-person sample with no control group, or tested on 8 kHz audio?).

Phase 3: Data Extraction & Mapping

PHASE OBJECTIVE

Extract the specific data points required by the project brief to populate the systematic mapping tables.

1. **Design the Extraction Table:** Set up a shared spreadsheet with the exact columns mandated by the brief:
 - Languages covered, hours of audio, number of speakers.
 - Dialect coverage (were regional dialects like Ekiti Yoruba or Wawa Igbo included, or was it quietly standardized?).
 - Recording conditions and sampling rate (e.g., 8 kHz vs 44.1 kHz).
 - Domain (clinical vs. general).
 - License and public availability.
 - Reported metrics (WER, MOS, or consequence-weighted metrics).
 - Baselines compared against.
 - Telephony tested (Yes/No/Partial).
 - Code-switching handled (Yes/No).
2. **Data Extraction:**
 - Populate the table for every included paper, dataset (e.g., NaijaVoices, AfriSpeech-200, YORULECT), and model (e.g., Whisper, Meta MMS, XLS-R).
 - Identify the “Empty Cells”: Explicitly log where data is missing (e.g., lack of medical vocabulary pronunciation, absence of Pidgin resources).

RULE CHECK

If a dataset claims dialectal coverage, consult a native speaker to verify the claim; do not trust the dataset card alone.

Phase 4: Optional Empirical Experiment

PHASE OBJECTIVE

This step requires prior approval from management and a timeline extension. If approved, move beyond mapping to empirically prove the telephony degradation point.

1. **Setup:** Request the necessary budget/compute from Mariam for an audio processing setup.
2. **Execution:** Take existing high-quality benchmark audio for Nigerian languages and apply an 8 kHz codec simulation (mimicking a basic phone call).
3. **Testing:** Run the simulated narrowband audio through flagship ASR models (Whisper, MMS) and record the drop in accuracy/WER compared to their performance on the original clean audio.

Phase 5: Synthesis & Manuscript Writing

PHASE OBJECTIVE

Draft the paper targeting high-tier venues (LREC, INTERSPEECH, ACL workshops) while maintaining absolute scientific independence from the Mama AI product goals.

1. **Generate the PRISMA-ScR Flow Diagram:** Visually map the journey from the initial number of search results to the final number of included papers.
2. **Draft the Paper Structure:**
 - *Abstract (200-250 words):* Define the problem, the map, and the blank spaces.
 - *Introduction:* The healthcare context and the need for this map.
 - *Background:* Define ASR, TTS, WER, narrowband, etc.
 - *Methodology:* Detail the OSF protocol, databases, strings, and screening process.
 - *The Map (Tables):* Present the extracted data.
 - *Thematic Findings:* Synthesize by theme (Data availability, evaluation practice, dialectal coverage, deployment realism), *not* by listing one paper per paragraph.
 - *Gaps & Implications:* Highlight the empty cells (e.g., the lack of 8 kHz testing, the lack of clinical vocabulary).

CLAIMS DISCIPLINE CHECK

- Ensure the draft does not conflate "acceptability" with "effectiveness."
- Ensure it does not equate "benchmark accuracy" with "clinical safety."
- Do not shy away from the Kilkari precedent: if the evidence suggests the Mama AI architecture is currently unsafe or unsupported by the literature, state it plainly.

3. **Review and Finalize References:** Verify every citation and ensure APA 7th formatting.

Phase 6: Submission & Logistics

1. **Final Review:** Have the team review the paper specifically against the instructions in the initial Workstream A brief.
2. **Target Venue Selection:** Confirm the target venue (e.g., INTERSPEECH or LREC). If open-access fees are required, raise this with Mariam *before* submission.
3. **Submit the Paper.**

Appendix: Evaluation Checklists for Papers found

The Mapping Table Extraction (Hard Data)

- **Languages & Dialects:** Which exact languages are covered? Did they capture authentic regional dialects (e.g., Wawa Igbo, Ekiti Yorùbá, Sokoto Hausa) or did they quietly standardize the speech?
- **Scale:** What are the total hours of audio and the number of distinct human speakers?
- **Recording Conditions:** Was it recorded in a clean studio environment (16–48 kHz) or subjected to real-world telephony degradation (8 kHz)?
- **Domain:** Is the audio general conversation, news reading, or specific clinical/medical vocabulary?
- **Tasks Handled:** Does the model explicitly test for code-switching or language identification?
- **Metrics & Baselines:** What exactly was measured (e.g., WER, MOS)? What previous model did they use as a baseline to prove their new version is better?
- **Availability:** Is the dataset or model publicly available to researchers, and under what specific license?

Methodological Red Flags (Quality Assessment)

- **Data Leakage:** Is there any risk that their training data accidentally bled into their test set, artificially inflating their score?
- **Class Imbalance:** Is standard, textbook speech padding their overall accuracy score while minority regional accents are failing?
- **External Validation:** Did they test their model on an independently gathered dataset, or only on their own comfortable, held-out test split?
- **Control Group:** If it was a field test, was there a proper control group for comparison, or was it just an isolated trial on a small group?
- **Metric Appropriateness:** Are they over-relying on basic accuracy instead of analyzing the real-world severity of the errors?
- **Reproducibility & Theory:** Did they publish enough detail for someone else to replicate it, and is the work grounded in actual clinical or linguistic theory?

The Mama AI Reality Check (Claims Discipline)

- **Transferability:** Will these findings actually hold up in Nigeria over a basic 2G/3G cellular network?
- **Acceptability vs. Effectiveness:** Are the authors claiming the tool “improves health outcomes” just because a group of users reported high satisfaction scores?
- **Benchmark vs. Safety:** Does the paper assume that just because the AI aced a controlled computer test, it’s automatically safe to use on real patients in medical emergencies?

When you find an AI Model (The “Brain”)

An AI model is the actual software program that listens to or speaks the language.

- **What is its main job?** Is it trying to turn voice into written text, turn written text into a voice, or just guess which language is being spoken?
- **Can it handle bad phone connections?** Did the researchers test the AI on muffled, low-quality phone audio (8 kHz), or did they only test it on crystal-clear studio recordings? (*This is crucial for the Mama AI project*).
- **How do they score mistakes?** Do they count every mistake equally? Or do they grade the AI more harshly if it mishears a dangerous medical word (like “bleeding”) compared to a harmless word (like “the”)?
- **Does it understand mixed languages?** Can it keep up if a speaker naturally switches back and forth between English and a local language in the same sentence?
- **Who are they comparing it to?** To prove this AI is actually good, what older or standard AI did they test it against?
- **Can anyone use it?** Is the software free and open for our engineers to download, or is it a paid product hidden behind a tech company’s wall?

When you find a Dataset (The “Training Material”)

A dataset is the massive collection of audio recordings used to teach the AI.

- **Which languages and accents are *actually* in it?** Did they capture real regional accents (like rural dialects), or did they only record people speaking the formal, “textbook” version of the language?
- **How big is it?** Note the total hours of audio and exactly how many different

human beings were recorded. (More people = better AI).

- **How was it recorded?** Were people speaking into expensive microphones in quiet rooms, or using normal cell phones with real-world background noise?
- **What are the people talking about?** Are they reading random news articles and Bible verses, or are they specifically using medical and healthcare words?
- **Are we legally allowed to use it?** Check the license. Is it completely free for our project to use, or do we need special permission?
- **Did they cheat the test?** (*Look for "Data Leakage"*). Is there any evidence that they accidentally tested the AI using the exact same audio clips they used to teach it? (This is like giving a student the answers before a test).